Built a report with R



Written by Nurlaila Tahawan นิ๊ว Porfolio

(●'◡'●) Hello everyone

Today, we learn about how to make R Markdown
and build a report,
Details are made for 4 charges.

I learned it from "Data Science Bootcamp Batch 12" 👈
And I'm sharing it with you guys!

Do you know R Markdown?
R Markdown is for making reports in files
ex.
- PDF
- HTML
- Docs
- Others

It's not difficult, Guys!
You can make it in RStudio (Posit)

And you can use rmarkdown.PDF for guide liene in your project.

And this's our fun topic today!

1. Free Datasets
2. Simple to use Posit. cloud to Built a report with R
    1. Open Posit. cloud website
    2. Select "New studio project."
    3. Create a new R markdown document
    4. Make Title and select HTML or PDF
    5. Multi-language Code Chunks
    6. Knit the current document (Ctrl+Shift+K)
    7. message=FALSE, warning=FALSE
3. Data analysis using dplyr a package
4. This is my final work as a PDF
5. Line-by-Line Breakdown
    1. Fetching data with dplyr
    2. The Density Plot (ggplot2)
    3. Making a colorful Scatter Plot
    4. Splitting dimensions with Facet Wrap
    5. Checking price spreads with a Violin Plot

Free Datasets

Guys... are you looking for several Free Datasets for practice to make data analysis? You can find on this website

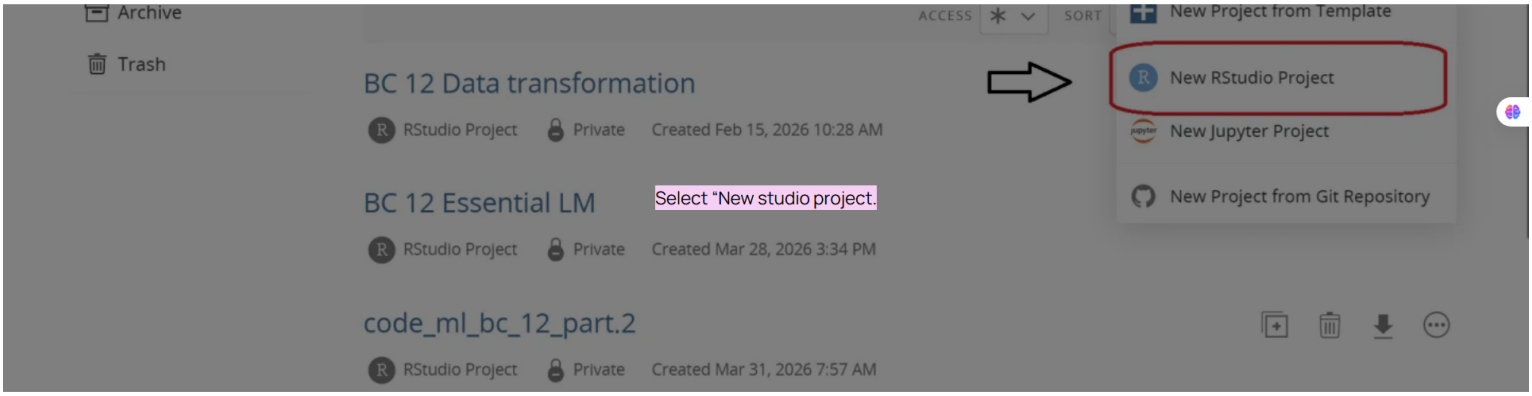
- Kaggle Datasets
- data.world

Simple to use Posit. cloud to Built a report

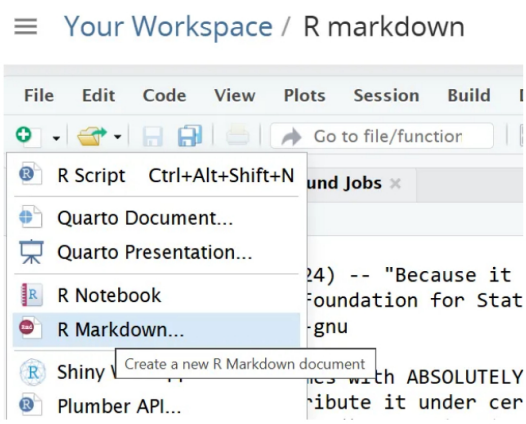
with R

Open Posit. cloud website

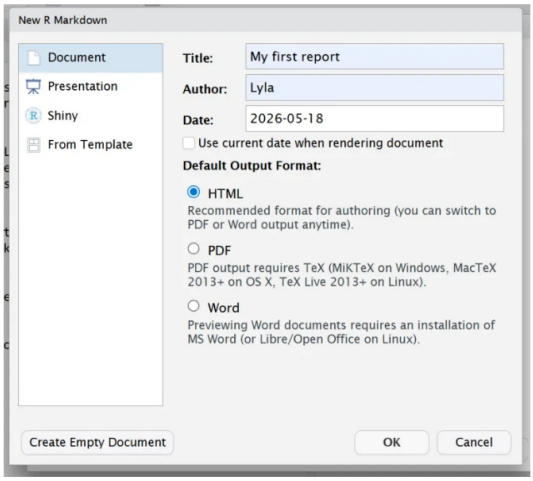
Select "New studio project."



Create a new R markdown document



Make Title and select HTML or PDF





Multi-language Code Chunks

Clear any data without your contant and run your code!

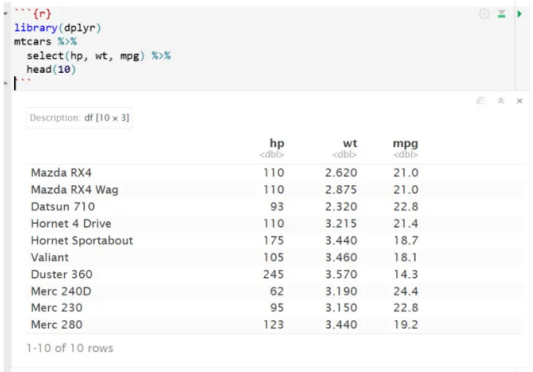
- Select "code chunk symbols" to make a blog "Code Chunks"



- This is Multi-language Code Chunks

```
1 ```{}
2
3 ```
```





```{r}
library(dplyr)
mtcars %>%
  select(hp, wt, mpg) %>%
  head(10)
```

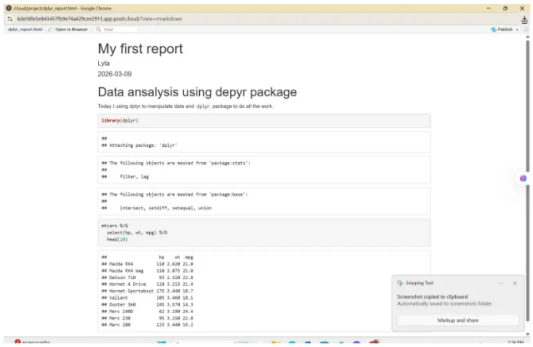
Description: df [10 × 3]

	hp <dbl>	wt <dbl>	mpg <dbl>
Mazda RX4	110	2.620	21.0
Mazda RX4 Wag	110	2.875	21.0
Datsun 710	93	2.320	22.8
Hornet 4 Drive	110	3.215	21.4
Hornet Sportabout	175	3.440	18.7
Valiant	105	3.460	18.1
Duster 360	245	3.570	14.3
Merc 240D	62	3.190	24.4
Merc 230	95	3.150	22.8
Merc 280	123	3.440	19.2

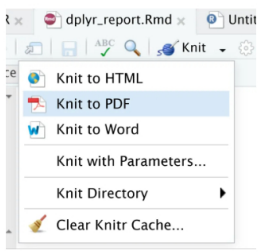
1-10 of 10 rows

Knit the current document (Ctrl+Shift+K)

When you finish your code project, and you want to show your Console. You can use (Ctrl+Shift+K).



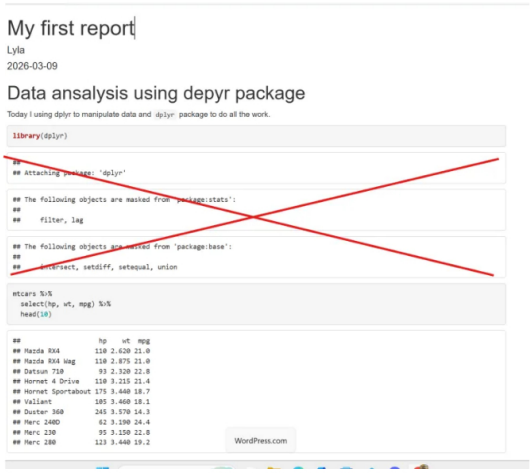
Just one **Knit** click. or **(Ctrl+Shift+K).** So this is the Console.



Or, if you want to make a PDF or others, you can select it easily like this.

message=FALSE, warning=FALSE

If you don't want to show your Message on your Console.



So, put ***message=FALSE, warning=FALSE*** In the code, when you want to clean your Message on the Console.

```
message=FALSE, warning=FALSE
```

For example

```
```{r message=FALSE, warning=FALSE}
ggplot(diamonds %>%
         sample_n(200),
       aes(x = cut,
           y = price))+
  geom_violin()
```
```

Run the code, and you can see the difference like this.

My first report

Lyla

2026-03-09

Data ansalysis using depyr package

Today I using dplyr to manipulate data and `dplyr` package to do all the work.

```
library(dplyr)
mtcars %>%
  select(hp, wt, mpg) %>%
  head(10)
```

```
##                   hp    wt  mpg
## Mazda RX4         110 2.620 21.0
## Mazda RX4 Wag     110 2.875 21.0
## Datsun 710         93 2.320 22.8
## Hornet 4 Drive    110 3.215 21.4
## Hornet Sportabout 175 3.440 18.7
## Valiant           105 3.460 18.1
## Duster 360        245 3.570 14.3
## Merc 240D          62 3.190 24.4
## Merc 230           95 3.150 22.8
## Merc 280          123 3.440 19.2
```

This is the end of the report.

Well done yo,u can make it now.

Data analysis using `dplyr` a package

```
---
title: "My first report"
author: "Lyla"
date: "2026-03-09"
output:
  pdf_document: default
  html_document: default
```

````
---

# Data analysis using "depyr" package
Today, I am using dplyr to manipulate
data and the `dply` package to do all the work.

```{r message=FALSE, warning=FALSE}
library(dplyr)
mtcars %>%
  select(hp, wt, mpg) %>%
  head(10)

```

## create a histogram
```{r message=FALSE, warning=FALSE}
library(ggplot2)
ggplot(mtcars, aes(x = mpg)) +
  geom_density() +
  theme_minimal()
```

```{r message=FALSE, warning=FALSE}
library(ggplot2)
ggplot(diamonds %>%
         sample_n(300),
       aes(x = carat,
           y = price,
          color = cut))+
  geom_point(alpha = 0.8 , size = 1.5 )+
  scale_color_brewer(type = "qual", palette = 6)+
  theme_minimal()

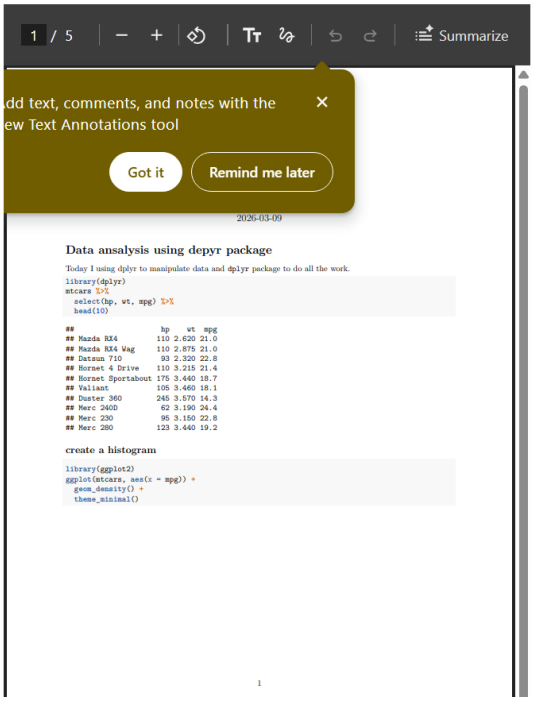
```
```{r message=FALSE, warning=FALSE}
set.seed(42)
ggplot(diamonds %>%
         sample_n(1000),
       aes(x = carat,
           y = price,
           alpha = 0.5))+
  geom_point(alpha = 0.2, color = "salmon")+
  facet_wrap(~cut, ncol= 5)+
  theme_minimal()

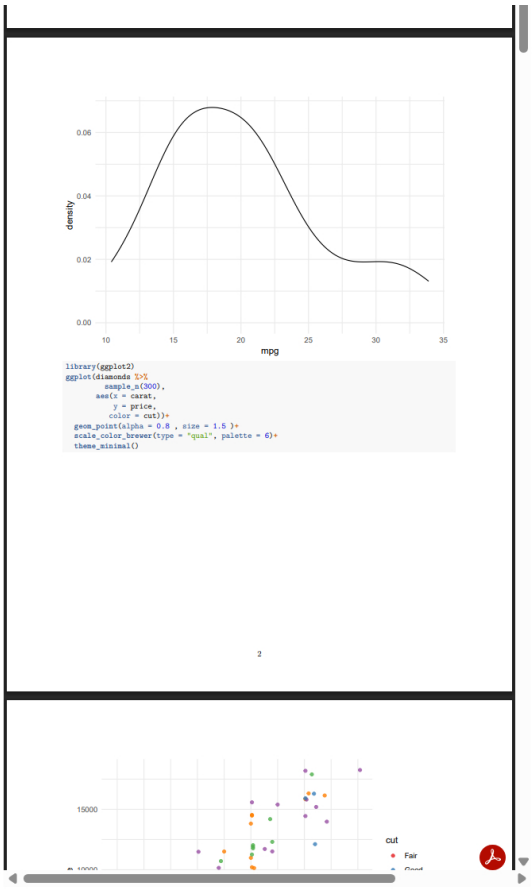
```

```{r message=FALSE, warning=FALSE}
ggplot(diamonds %>%
         sample_n(200),
       aes(x = cut,
           y = price))+
  geom_violin()
```

This is the end of the report.
````

This is my final work as a PDF





R-markdown report ดาวน์โหลด

This is the end of the report.

Line-by-Line Breakdown

To explain the details of the code.

Fetching data with `dplyr`

```
# Fetching with dplyr
library(dplyr)
mtcars %>%
  select(hp, wt, mpg) %>%
  head(10)
```

อธิบาย

`mtcars %>%`: Grab the built-in car spreadsheet... and then pass it down the conveyor belt!

`library(dplyr)`: Opening up your data-manipulation toolbox.

`select(hp, wt, mpg)`: Cherry-pick only three columns: Horsepower, Weight, and Miles Per Gallon.

`head(10)`: Just show me the top 10 rows so my screen doesn't explode with data.

Description: df [10 × 3]

	hp <dbl>	wt <dbl>	mpg <dbl>
Mazda RX4	110	2.620	21.0
Mazda RX4 Wag	110	2.875	21.0
Datsun 710	93	2.320	22.8
Hornet 4 Drive	110	3.215	21.4
Hornet Sportabout	175	3.440	18.7
Valiant	105	3.460	18.1
Duster 360	245	3.570	14.3
Merc 240D	62	3.190	24.4
Merc 230	95	3.150	22.8
Merc 280	123	3.440	19.2

1-10 of 10 rows

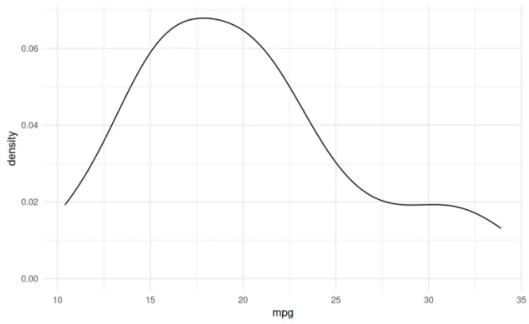
The Density Plot (ggplot2)

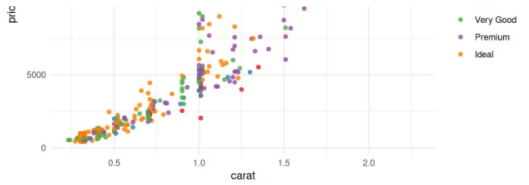
```
# 2. The Density Plot (ggplot2)
ggplot(mtcars, aes(x = mpg)) +
  geom_density() +
  theme_minimal()
```

`ggplot(mtcars, aes(x = mpg))`: Tell R we are using the car spreadsheet and mapping **`mpg`** (continuous data) to the X-axis.

`+ geom_density()`: Add a layer that draws a smooth, wave-like hill showing where most cars' fuel efficiency clusters.

`+ theme_minimal()`: Slap on a clean, modern white background.



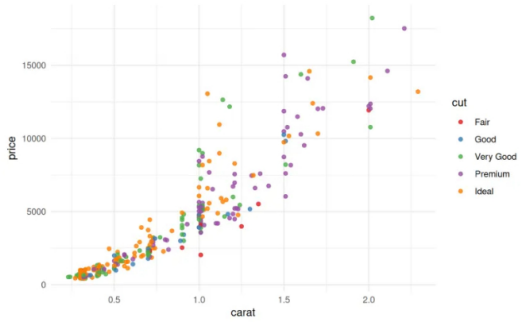


Splitting dimensions with Facet Wrap

```
# 4. Splitting dimensions with Facet Wrap
set.seed(42)
ggplot(diamonds %>% sample_n(1000),
       aes(x = carat, y = price, alpha = 0.5)) +
  geom_point(alpha = 0.2, color = "salmon") +
  facet_wrap(~cut, ncol= 5) +
  theme_minimal()
```

be slightly transparent (**`alpha`**) and a specific size.

`scale_color_brewer(...)`: Swap out the default colors for a gorgeous, professionally curated color palette.

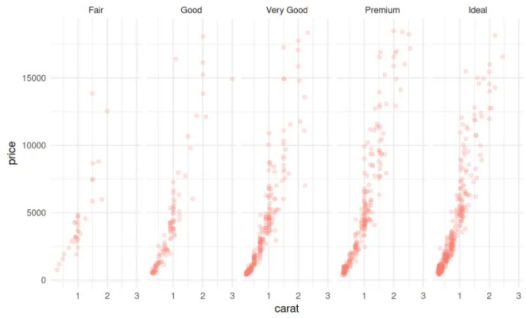


Splitting dimensions with Facet Wrap

```
# 4. Splitting dimensions with Facet Wrap
set.seed(42)
ggplot(diamonds %>% sample_n(1000),
       aes(x = carat, y = price, alpha = 0.5)) +
  geom_point(alpha = 0.2, color = "salmon") +
  facet_wrap(~cut, ncol= 5) +
  theme_minimal()
```

`set.seed(42)`: The pseudo-random "freeze ray"! It ensures that every time you run this code, you get the *exact same* 1,000 random rows.

`facet_wrap(~cut, ncol = 5)`: The Ultimate Power Move! Instead of crowding all 1,000 dots onto one messy graph, this smashes it into 5 distinct mini-charts side-by-side, one for each diamond cut type..



Checking price spreads with a Violin Plot

```r
# 5.Checking price spreads with a Violin Plot
```{r message=FALSE, warning=FALSE}
ggplot(diamonds %>%
         sample_n(200),
       aes(x = cut,
           y = price)) +
  geom_violin()
```

`ggplot(` — Tells R: *"Hey hun, let's start a brand new blank canvas chart!"*

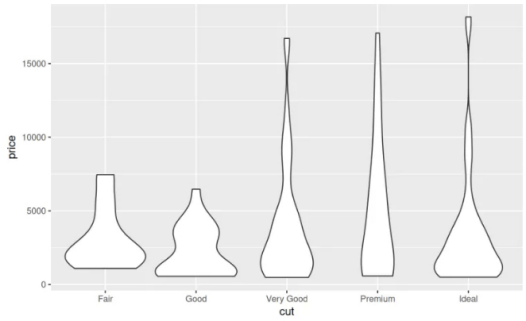
`diamonds %>%` — Grabs your diamond spreadsheet and then passes it down the conveyor belt.

`sample_n(200),` — Randomly chops the data down to just 200 rows. Super fast!

`aes(x = cut, y = price))` — Mapping time! Puts the `cut` (words/discrete data) on the X-axis and `price` (decimals/continuous data) on the Y-axis.

`+` — The magic bridge! Tells R: *"Keep reading downstairs, I'm adding a layer!"*

`geom_violin()` — Squirt! It draws smooth curves. The wider the belly, the more diamonds live at that price point.



"And that's a wrap on Built a report with R! Give it a spin and have fun with it. Huge props to everyone for making it this far—you rock! Wishing you all pure health, sharp minds, and great vibes. Stay awesome!"


```



Have a good day!

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